

SupplyGuard: Evaluating Commodity-Specific Adaptive Conformal Calibration for Agricultural Market-Arrival Forecasting

M.Tech Research Project

Rough draft for staff review; author and affiliation to be completed

Abstract—On the later April-June benchmark, Reselected LGBM bundle has the lowest WAPE among the recorded main and preserved-reference forecasts (19.80%). For identical log-LightGBM point predictions, commodity fixed coverage is 90.79% with mean width 194.135 tonnes; commodity adaptive coverage is 94.88% with width 344.446 tonnes. Their mean interval scores are 495.119 and 475.210, respectively. These are empirical tradeoffs, not a universal superiority or causal claim. The later-period interval-score difference has a descriptive paired interval that includes zero. A later contiguous April-June 2024 cohort was newly acquired only after model and calibration settings were frozen. We compare pooled fixed, commodity fixed and rolling commodity adaptive calibration under a common scaled residual score and 95% nominal target. The benchmark includes rolling and weekly baselines, Random Forest, log-LightGBM and a validation-selected LightGBM bundle, with controlled price and calendar feature ablations. Findings are empirical and limited to the recorded cohort. No new conformal theorem, verified novelty or operational causal benefit is claimed.

Index Terms—Conformal calibration, agricultural market arrivals, time-series forecasting, uncertainty evaluation

I. Introduction and Research Question

Reliable planning requires uncertainty information as conditions change. A single point forecast does not express the range of plausible future arrivals. This study asks whether commodity-specific rolling adaptive calibration improves the coverage-width tradeoff relative to pooled and commodity-specific fixed calibration. We treat this as an empirical question and candidate domain contribution, not established methodological novelty. Market arrivals represent supply inflows, not retail demand or warehouse inventory.

II. Related Work

Gibbs and Candes introduce adaptive conformal inference for changing distributions, evaluated on financial volatility and election-night prediction [1]. We reuse its level-adjustment idea but add clipped, date-batched updates and residual windows; original guarantees do not automatically transfer. Romano et al. study group-conditional coverage [2]; commodity grouping here is an application of existing group calibration, not an invention. Xu and Xie’s EnbPI addresses dynamic time-series intervals [3]; EnbPI is related work and is not an implemented comparator in this draft. The M5 retail forecasting competition highlights boosted-tree and combination methods

[4]. Its sales targets and 28-day horizon differ from wholesale one-day market inflows. Our contribution is reproducible engineering and empirical evaluation; novelty beyond this remains uncertain.

III. Dataset Provenance

The completed government AGMARKNET cohort comprises 59,648 actual market-day observations in 16 Uttar Pradesh markets during 2021–2023 for rice, wheat, onion and potato. There are 144 cached monthly source responses with request parameters and SHA-256 hashes. Published daily total arrivals are counted once, avoiding repeated totals across varieties. Modal prices are arrivals-weighted across valid varieties, with median fallback for zero usable weights. Prices are INR per quintal and arrivals are tonnes. Market selection uses January 2021–June 2022 completeness only. Daily reindexing creates 10,374 unobserved calendar rows; targets and prices remain missing and are not replaced by zero. Another 4,965 January–March 2024 observations were previously evaluated. A later contiguous April–June 2024 cohort was newly acquired only after model and calibration settings were frozen. Source provenance and subgroup missingness are retained in machine-readable manifests. The larger four-state plan was not completed because of acquisition restrictions; conclusions cover one state. Historical acquisition in 2026 does not make the records current 2026 market observations.

IV. Methodology

A. Point Forecasting

We compare rolling mean, weekly seasonal mean, RandomForestRegressor, log-target LightGBM and a commodity-routed LightGBM bundle. Eight LightGBM specifications are configurations of one model family, not eight algorithms. Random Forest compares raw and log targets with 120 trees, depth 18, minimum leaf size 3 and maximum feature fraction 0.7. Its numeric median imputer and one-hot category encoder fit only on proper training data. LightGBM uses native missing-value and category handling. All receive the same source information and eligible target rows; representation differs. Log targets are inverted with `expm1`, ratio targets are rescaled with the past mean, and predictions are clipped below at zero.

B. Feature Information

The 34 inputs contain calendar-day arrival lags at 1, 2, 3, 7, 14, 21 and 28 days; past rolling mean and standard deviation; prior price; 7/14/28-day means, medians and standard deviations; weekly seasonal lag average; 7/28-day price means; trend and recent ratios; weekday, month and day-of-year; sine/cosine weekday and annual seasonality; and training-frozen state, market and commodity codes. All rolling features shift outcomes by one day before calculation. Unknown vocabulary values encode as -1; Random Forest ignores unknown one-hot categories. Fuel, warehouse stock, strikes and floods are not model inputs.

C. Common Calibration Score

For actual arrival y_i , point prediction $\hat{y}_i$ and past-only scale $s_i = \max(\bar{y}_{i,28}, 1)$, define

$$r_i = |y_i - \hat{y}_i| / s_i. \quad (1)$$

The fixed split quantile uses order rank $\lceil (n+1)(1-\alpha) \rceil$ with $\alpha = 0.05$. Intervals are $[\max(0, \hat{y}_i - q_i s_i), \hat{y}_i + q_i s_i]$. Pooled fixed uses all calibration residuals; commodity fixed uses crop-specific residuals. The existing state/commodity grouping reduces to commodity grouping in this one-state cohort and is reused transparently.

D. Date-Batched Adaptive Calibration

Commodity adaptive calibration retains the last W residual observations per crop. The validation grid uses $W \in \{250, 500\}$ and $\gamma \in \{0.005, 0.02\}$. All observations in date batch t receive intervals before any outcomes in that batch update the histories. After revelation,

$$\alpha_{c,t+1} = \text{clip}(\alpha_{c,t} + \gamma(0.05 - \bar{m}_{c,t}), 0.005, 0.2), \quad (2)$$

where $\bar{m}$ is the crop's batch miss rate. Fewer than 100 group residuals or an unsupported order rank triggers fallback to the original pooled 95% quantile. Clipping and fallback counts are recorded. The rolling, clipped and batched implementation is ACI-inspired; no general coverage guarantee is claimed.

V. Experimental Protocol

The task is rolling one-day-ahead prediction using observations strictly before each target date. It assumes earlier reporting is available and does not evaluate multi-day forecasts from one frozen origin. Three expanding validation quarters in 2023 each use a preceding 90-day calibration segment withheld from that fold's proper training data. All learned configurations use identical partitions. Model selection minimizes mean validation WAPE plus 0.25 times its across-fold standard deviation; crop routing is selected per commodity. Random Forest and log-LightGBM choices use global validation aggregates. Adaptive settings minimize validation mean interval score among configurations with mean coverage at least 94%; otherwise closest nominal coverage is chosen, then interval score. Final proper training ends September 2023,

with October–December calibration. Settings and model hashes freeze before later acquisition. January–March 2024 is explicitly previously inspected. Later evaluation starts adaptive histories from the same October–December pool, not January–March residuals; prior observed arrivals remain available as forecasting features.

A. Metrics and Ablations

Point metrics are MAE and RMSE in tonnes, WAPE and R^2 . WAPE is an error measure, not accuracy. Interval metrics are empirical coverage, signed gap from 95%, mean width and mean interval score:

$$IS_\alpha(l, u; y) = (u - l) + \frac{2}{\alpha}(l - y)\mathbf{1}\{y < l\} + \frac{2}{\alpha}(y - u)\mathbf{1}\{y > u\}. \quad (3)$$

A prespecified log-LightGBM (15 leaves, 400 trees) is retrained with full features, without price_lag/price_mean_7/price_mean_28, and without weekday/month/year_day plus four cyclic features. Arrival lags still retain indirect seasonality. Rows and remaining settings stay identical. Fixed commodity calibration is the primary ablation comparison; inherited adaptive settings are secondary. Results are saved overall and by crop, market and month, with at least 30 subgroup observations. Paired differences use 300 resamples of entire dates, retaining all markets within each date. This preserves within-date dependence but treats date blocks as independent and ignores serial dependence. Bootstrap intervals are descriptive; an interval-score difference whose interval includes zero does not establish clear superiority. Training and inference times are measured on one machine, excluding common feature construction, and are not speed guarantees.

VI. Executed Results

On the later April–June benchmark, Reselected LGBM bundle has the lowest WAPE among the recorded main and preserved-reference forecasts (19.80%). For identical log-LightGBM point predictions, commodity fixed coverage is 90.79% with mean width 194.135 tonnes; commodity adaptive coverage is 94.88% with width 344.446 tonnes. Their mean interval scores are 495.119 and 475.210, respectively. These are empirical tradeoffs, not a universal superiority or causal claim. The later-period interval-score difference has a descriptive paired interval that includes zero.

A. Previously inspected January–March 2024

The evaluation contains 4,965 eligible observed targets. Missing calendar rows are excluded rather than imputed as zero. adaptive-minus-pooled_fixed: coverage difference 0.0270 [0.0213, 0.0331]; width difference 45.6996 [42.3307, 49.5597]; interval_score difference -17.0553 [-34.9816, 0.4542]. adaptive-minus-commodity_fixed: coverage difference 0.0278 [0.0226, 0.0336]; width difference

TABLE I
Point forecasts: Previously inspected January-March 2024.

Model	MAE	RMSE	WAPE	R ²
Rolling mean	37.960	150.312	27.71%	0.644
Weekly seasonal	43.493	161.742	31.75%	0.588
Random Forest	31.970	149.687	23.34%	0.647
Log-LightGBM	29.810	149.830	21.76%	0.646
Reselected LGBM bundle	30.945	154.259	22.59%	0.625
Original log-LGBM / full	29.946	149.437	21.86%	0.648
Existing LGBM bundle	31.242	155.374	22.81%	0.620

TABLE II
Same point forecasts, 95% nominal intervals. Gap is in percentage points.

Calibration	Cov.	Gap	Width	IS
Commodity adaptive	94.72%	-0.28	194.471	374.341
Commodity fixed	91.94%	-3.06	148.822	392.924
Pooled fixed	92.02%	-2.98	148.772	391.396

TABLE III
Commodity breakdown: Previously inspected January-March 2024.

Crop	Method	n	Cov.	Width
Onion	Crop adaptive	1288	94.80%	70.807
Potato	Crop adaptive	1289	94.80%	178.090
Rice	Crop adaptive	1193	94.22%	242.538
Wheat	Crop adaptive	1195	95.06%	297.443
Onion	Crop fixed	1288	93.17%	60.555
Potato	Crop fixed	1289	93.72%	160.762
Rice	Crop fixed	1193	89.94%	169.505
Wheat	Crop fixed	1195	90.71%	210.429
Onion	Pooled fixed	1288	92.78%	58.200
Potato	Pooled fixed	1289	93.87%	165.922
Rice	Pooled fixed	1193	91.03%	179.894
Wheat	Pooled fixed	1195	90.21%	196.822

TABLE IV
Feature ablations: same rows and fixed log-LightGBM settings.

Variant	MAE	WAPE	Cov.	Width
Without price	30.031	21.93%	92.06%	150.912
Without calendar	29.747	21.72%	92.23%	151.583
Original log-LGBM / full	29.946	21.86%	91.92%	150.124

45.6498 [42.3252, 49.5678]; interval_score difference - 18.5835 [-33.4814, -3.7106].

B. Later April-June 2024

The evaluation contains 4,801 eligible observed targets. Missing calendar rows are excluded rather than imputed as zero. adaptive-minus-pooled_fixed: coverage difference 0.0423 [0.0357, 0.0490]; width difference 155.9624 [133.0650, 183.4572]; interval_score difference -25.2021 [-87.9286, 34.5279]. adaptive-minus-commodity_fixed: coverage difference 0.0408 [0.0350, 0.0486]; width difference 150.3115 [127.4359, 173.7172]; interval_score difference -19.9093 [-76.6260, 36.1274].

TABLE V
Point forecasts: Later April-June 2024.

Model	MAE	RMSE	WAPE	R ²
Rolling mean	63.317	179.693	33.51%	0.728
Weekly seasonal	72.675	203.804	38.46%	0.650
Random Forest	38.712	112.172	20.49%	0.894
Log-LightGBM	38.272	118.213	20.25%	0.882
Reselected LGBM bundle	37.424	116.869	19.80%	0.885
Original log-LGBM / full	38.538	120.909	20.39%	0.877
Existing LGBM bundle	37.977	117.973	20.10%	0.883

TABLE VI
Same point forecasts, 95% nominal intervals. Gap is in percentage points.

Calibration	Cov.	Gap	Width	IS
Commodity adaptive	94.88%	-0.12	344.446	475.210
Commodity fixed	90.79%	-4.21	194.135	495.119
Pooled fixed	90.65%	-4.35	188.484	500.412

TABLE VII
Commodity breakdown: Later April-June 2024.

Crop	Method	n	Cov.	Width
Onion	Crop adaptive	1253	95.37%	75.904
Potato	Crop adaptive	1254	95.30%	154.403
Rice	Crop adaptive	1147	93.98%	144.153
Wheat	Crop adaptive	1147	94.77%	1045.870
Onion	Crop fixed	1253	91.94%	55.954
Potato	Crop fixed	1254	93.86%	129.672
Rice	Crop fixed	1147	91.19%	109.278
Wheat	Crop fixed	1147	85.79%	500.418
Onion	Pooled fixed	1253	91.22%	53.782
Potato	Pooled fixed	1254	94.34%	133.821
Rice	Pooled fixed	1147	92.07%	115.908
Wheat	Pooled fixed	1147	84.57%	467.971

TABLE VIII
Feature ablations: same rows and fixed log-LightGBM settings.

Variant	MAE	WAPE	Cov.	Width
Without price	38.576	20.41%	90.84%	194.975
Without calendar	38.664	20.46%	91.06%	199.220
Original log-LGBM / full	38.538	20.39%	90.92%	194.067

VII. Discussion and Limitations

Coverage and width must be interpreted together. On the later period, adaptive calibration improves coverage by widening intervals substantially; its mean interval-score reduction has a date-block interval crossing zero. This result supports a tradeoff, not clear dominance. A narrower interval can arise from undercoverage, and lower aggregate error can hide commodity or market weaknesses. A validation winner need not remain a test winner. The inspected quarter cannot provide a new independent generalization claim. Even a newly acquired later period is limited to the same region and reporting process. Windows and model searches are small and budget-limited. Reporting latency, calendar dependence

and group sparsity remain important practical issues. Independent date-block resampling understates some temporal dependence uncertainty.

A. Supporting Components

The webpage’s main and regional warehouse inventory, consumption, fuel prices, strikes and floods are deterministic synthetic scenarios. Its coloured response scores are heuristic feasibility assessments, not calibrated success probabilities. Delhivery logistics records come from an educational mirror and support separate exploratory observational analysis; there is no verified shipment join to market data. Causal identification remains unsupported, and real actions stay withheld. Existing predictive gate ablations accepted 100% of historical predictions and demonstrate no selective benefit. No demand forecasting, stock-out reduction, savings, disaster prediction or causal improvement is inferred from market-arrival metrics.

B. Unfinished Work

This is a rough staff-review draft, not a publication-ready paper. Author/affiliation, broader literature coverage, external-region validation, reporting-delay audit, serial-dependence-aware inference, additional time-series calibration comparators and prospective business evaluation remain unfinished. We do not claim first use, novelty, a new theorem or universal superiority.

VIII. Conclusion

SupplyGuard provides an executed, reproducible comparison of fixed pooled, fixed commodity and rolling adaptive commodity calibration around common one-day-ahead agricultural market forecasts. The findings should be read as measured coverage-width-score trade-offs within the documented cohort. Negative and inconclusive outcomes remain part of the record. The working application and earlier model artifacts are preserved separately.

References

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